VMS System

Complete Workflow Scenarios

All 18 Major Workflows with Decision Points and Edge Cases

For Capstone Project Defense

Version 2.0 | July 2026

Workflows Covered (18 Complete Scenarios)

1. Staff Reports a Vehicle Defect
2. Log a Condition Check
3. Run a Readiness Check
4. Create a Maintenance Ticket
5. Custodian Inspects Main Issue
6. Assign Mechanic to Sub-Issue
7. Mechanic Logs Repair Work
8. Custodian Verifies Repair
9. Admin Confirms Repair
10. Defer a Sub-Issue
11. Close Ticket (Complete/Done)
12. Close Ticket (Decision Close)
13. Reassign Mechanic Mid-Repair
14. Create Preventive Maintenance Schedule
15. Complete Scheduled Maintenance
16. Add Maintenance Record (Standalone)
17. Decommission a Vehicle
18. Add Sub-Issue Mid-Ticket

1. Staff Reports a Vehicle Defect

A staff member discovers a problem with a vehicle (brake squeak, overheating, flat tire) and files an Issue Report. An Admin reviews and creates a Maintenance Ticket. The Custodian inspects, a Mechanic repairs, and the cycle closes with verification and confirmation.

SCENARIO: BRAKE PROBLEM DISCOVERED

Sarah (Custodian) Custodian notices Ambulance AMB-101's brakes feel soft during routine transport. Files Issue Report: type = Brake Problem, severity = High, description = "Brakes feel soft and unresponsive", timestamp = 2026-07-20 14:30.

Paul (Admin) Admin reviews pending Issue Reports queue. Sees Sarah's report, phones her to confirm the finding. She confirms brakes lack responsiveness. Paul approves the report and clicks "Create Ticket".

Paul (Admin) Admin creates Maintenance Ticket: vehicle=AMB-101, linked_issue_report=BR-2026-0412, description="Soft brakes - safety critical", assigns Sarah as Custodian for inspection, status=Open.

Sarah (Custodian) Custodian receives notification. Opens ticket, performs physical brake inspection: checks brake pad wear (visually worn), brake fluid level (low), hose integrity (good). Submits inspection verdict: Needs Maintenance, creates sub-issue: "Replace brake pads - wear/replacement".

Paul (Admin) Admin sees ticket now has verified Main Issue. Assigns James (Maintenance Personnel) as mechanic to sub-issue, status changes from Open to Active.

James (Maintenance Personnel) Mechanic receives work order. Orders brake pads (part #BP-2426), receives them next day, removes old pads, installs new pads, bleeds brake lines. Logs repair work: date_completed=2026-07-21, parts_used="Brake Pads

#BP-2426", cost=125.00, notes="Pads replaced per spec. Bleed air from lines. Test pressure 800 psi nominal." Marks sub-issue "ready for verification".

Sarah (Custodian) Custodian receives verification notification. Test-drives ambulance, applies brakes at speeds 10mph, 25mph, 50mph. All responses firm and proportional. Checks attestation: "I personally tested this vehicle and confirm repair is effective". Marks verdict: Approved, sub-issue status = For Confirmation.

Paul (Admin) Admin sees sub-issue awaiting final confirmation. Reviews Sarah's test notes, James's work log, and parts receipt. Everything validates. Clicks Confirm. System marks sub-issue Done.

System creates Maintenance Record (immutable): ticket_id=TK-2026-1847, vehicle=AMB-101, work="Replaced brake pads", mechanic=James, custodian=Sarah, date_completed=2026-07-21, cost=125.00. Closes ticket (status=Closed, result=Done). Updates vehicle status to Ready. Marks Issue Report Resolved. Sends notifications to all three users.

EXPECTED OUTCOMES

- ✓ Ticket transitions: Open → Active → Closed (Done)
- ✓ Sub-issue transitions: Open → Under Repair → For Inspection → For Confirmation → Done
- ✓ Vehicle status: Available → Under Maintenance → Available
- ✓ Vehicle condition remains unchanged (Good)
- ✓ Issue Report status: Pending → Resolved
- ✓ Maintenance Record created and locked (immutable)
- ✓ All three roles notified at key transitions

DATA FLOW & DECISIONS

Custodian discovers issue → files Issue Report with type, severity, description, photo

Admin reviews; decides Approve or Reject

Decision Point 1 If Rejected → Issue Report discarded, no ticket created. If Approved → Continue.

Admin creates Ticket, assigns Custodian, status = Open

Custodian inspects vehicle, submits verdict: No Issues OR Needs Maintenance

Decision Point 2 If No Issues → skip repair, ticket jumps to Admin confirmation and closes clean. If Needs Maintenance → create sub-issues.

Admin assigns each sub-issue to Mechanic, status = Active

Mechanic performs repair, logs work (cost, parts, date), marks sub-issue For Inspection

Custodian tests vehicle, submits verdict: Approved OR Rejected

Decision Point 3 If Rejected → sub-issue returns to Under Repair for rework. If Approved → continue.

Admin final confirmation: Approve (Done) OR Reopen (Under Repair)

System creates Maintenance Record, frees vehicle, closes ticket, resolves Issue Report

Edge Case 1: Inspection Finds No Defect
Sarah inspects and finds brakes are actually fine (false alarm). Submits "No Issues". Ticket skips repair entirely and goes to Admin confirmation. Ticket closes without work. Issue Report marked Resolved without repair work.

Edge Case 2: Mechanic Discovers Secondary Problem
While replacing brake pads, James discovers brake master cylinder also has a small leak. He completes the pads, notes the cylinder leak. Admin reviews and decides: (a) create new sub-issue on this ticket for master cylinder, or (b) defer to new ticket.

2. Log a Condition Check

Routine, comprehensive assessment of vehicle's physical condition (Good, Needs Maintenance, Not Fit for Service). Updates vehicle's condition field and status if needed. Non-expiring (stays current unless something changes).

SCENARIO: WEEKLY CONDITION INSPECTION - GOOD RESULT

Lisa (Custodian) Custodian performs weekly condition check on Firetruck FT-03. Inspects: engine (no leaks), suspension (good), brakes (responsive), lights (all work), body (no rust, no dents), hoses (intact). Logs Condition Check: result = Good, observations = "Routine weekly inspection. All systems nominal. No defects found."

System saves check, marks as Current for FT-03, updates vehicle.condition = Good. Vehicle status remains Available (unchanged).

Dashboard shows FT-03 condition = Good, displays "Current" badge on the inspection record.

SCENARIO: CONDITION CHECK FINDS DEFECT - NEEDS MAINTENANCE

Michael (Admin) Admin performs condition check on Transport Van TV-05. Finds: rust on frame (minor), suspension (worn), brakes (soft). Logs: result = Needs Maintenance, observations = "Frame rust, suspension components worn, brake pads approaching wear limit. Recommend repair soon."

System updates vehicle.condition = Needs Maintenance, updates vehicle.status = Under Maintenance (automatically). Creates History entry.

Paul (Admin) Admin reviews the condition check result. Sees vehicle is now Under Maintenance. Decides whether to create Issue Reports and Tickets for the identified problems, or monitor them.

EXPECTED OUTCOMES

- ✓ Condition Check record created with timestamp, result, observations
- ✓ Vehicle condition field updates to logged result
- ✓ If Needs Maintenance or Not Fit: vehicle status automatically changes to Under Maintenance
- ✓ If Good: vehicle status unchanged
- ✓ Check marked as "Current" in vehicle's condition history
- ✓ Previous check marked as "Past"
- ✓ Entry added to vehicle's audit history log
- ✓ Dashboard coverage % includes this vehicle

DATA FLOW & DECISIONS

Custodian / Admin opens Condition Monitoring, selects vehicle, inspects, enters result (Good/Needs Maintenance/Not Fit), observations

System saves check, updates vehicle.condition

Decision: Result Value

Result = Good → vehicle.status unchanged

Result = Needs Maintenance OR Not Fit → vehicle.status = Under Maintenance

Admin reviews result; if Needs Maintenance/Not Fit detected, may create Issue Reports and Tickets

Edge Case: Multiple Checks Same Vehicle
Lisa inspects Firetruck FT-03 on Monday (Good). On Friday, another inspector checks and finds Needs Maintenance. System creates two records. Friday check is marked "Current", Monday is "Past". Dashboard shows latest (Needs Maintenance). History log shows both entries with timestamps.

Edge Case: Not Fit for Service
David discovers Vehicle TV-12 has severe frame rust and structural damage. Logs result = Not Fit for Service. System automatically sets status = Under Maintenance. Vehicle cannot be dispatched. Admin may decide to decommission it.

3. Run a Readiness Check

Pre-deployment checklist specific to vehicle type. Each item Pass/Fail. Expires after 7 days. Vehicle can be Good condition but fail readiness (e.g., empty fuel tank). Required before dispatch.

SCENARIO: FIRETRUCK FAILS READINESS - EMPTY WATER TANK

Lisa (Custodian) Custodian before dispatch, runs readiness check on Firetruck FT-03. Loads type-specific checklist: Fuel Tank (full = Pass), Water Tank (half-empty = Fail), Pump operational (yes = Pass), All hoses aboard (yes = Pass), Lights & siren (all work = Pass), Equipment mounted (secured = Pass).

System detects one Fail (water tank). Overall readiness = Not Ready. Firetruck FT-03 cannot be dispatched until water tank is filled.

Lisa (Custodian) Custodian refills water tank to capacity. Runs readiness check again: all items Pass. Overall readiness = Ready.

System expires previous check (from 7+ days ago), marks new check as Current. FT-03 shows Ready. Vehicle can now be dispatched.

SCENARIO: CHECK PASSES - VEHICLE READY

Sarah (Custodian) Custodian before mission departure, runs readiness check on Ambulance AMB-101. All items (oxygen tank, medical kit, stretcher, defibrillator, communications, fuel, lights) = Pass. Logs check: result = Ready, expires_at = NOW + 7 days.

System marks AMB-101 as Ready. Vehicle can be dispatched immediately.

EXPECTED OUTCOMES

- ✓ Readiness Check record created with timestamp, item-by-item results
- ✓ Overall state: Ready (all Pass), Not Ready (any Fail), or Stale (no recent check)
- ✓ If Not Ready: vehicle cannot be dispatched (blocked by business logic)
- ✓ If Ready: vehicle can be dispatched
- ✓ Check expires after 7 days (configurable)
- ✓ Dashboard shows vehicle's ready/not-ready/stale badge
- ✓ Independent from Condition and Ticket status

DATA FLOW & DECISIONS

Custodian / Admin opens Vehicle → Readiness Check. System loads type-specific checklist.

User marks each item Pass or Fail, adds optional notes, submits

Calculation

If ALL items = Pass → state = Ready, expires at NOW + 7 days

If ANY item = Fail → state = Not Ready, no expiration (recheck required)

If most recent check > 7 days old → state = Stale (requires new check)

Edge Case: Good Condition, Failed Readiness

Ambulance AMB-105: Condition = Good (mechanically sound). Readiness = Not

Ready (oxygen tank empty). Cannot dispatch despite good condition. Exemplifies the separation of concerns — mechanical and operational readiness are independent.

Edge Case: Stale Check Expires Mid-Shift
Vehicle checked and Readiness = Ready on Day 1. On Day 8, staff tries to dispatch. Check is now Stale. System requires a new readiness check before dispatch, even though nothing may have changed. Safety-first design.

4. Create a Maintenance Ticket

Admin opens a formal work order on a vehicle. Assigns a Custodian to inspect the Main Issue. Can be created from Issue Report or standalone. Vehicle cannot have two open tickets on the same Main Issue simultaneously.

SCENARIO: CREATE TICKET FROM ISSUE REPORT

Paul (Admin) Admin opens Issue Reports queue, sees Sarah's "Brake Problem" report filed on 2026-07-20. Clicks "Create Ticket", selects Sarah as Custodian to verify the main issue.

System creates Maintenance Ticket: ID=TK-2026-1847, vehicle=AMB-101, status=Open, linked_issue_report=BR-2026-0412, custodian=Sarah, created_at=2026-07-20 15:00.

Sarah (Custodian) Custodian receives notification: "You've been assigned Maintenance Ticket TK-2026-1847: Brake Problem (AMB-101)". Opens ticket to view details and schedule inspection.

SCENARIO: CREATE TICKET STANDALONE (NO PRIOR ISSUE REPORT)

Paul (Admin) Admin opens Maintenance Tickets, clicks "Add New". Fills: vehicle=Firetruck FT-02, description="Engine oil change - preventive maintenance", priority=Medium, assigns Lisa as Custodian (even for preventive, maintenance requires oversight).

System creates ticket: ID=TK-2026-1848, status=Open, no linked_issue_report. Lisa receives notification.

EXPECTED OUTCOMES

- ✓ Ticket created with unique ID
- ✓ Status = Open (awaiting inspection)
- ✓ Custodian assigned and notified
- ✓ If from Issue Report: ticket links to report
- ✓ Vehicle status unchanged at ticket creation
- ✓ Ticket appears in Admin's queue and Custodian's assignment list

DATA FLOW & VALIDATION

Admin Maintenance Tickets → Add, or Issue Report → "Create Ticket"

Admin fills: vehicle, custodian, priority, description

Validation 1: Duplicate Check

Does vehicle have Open ticket with identical Main Issue? If YES → blocked with error "Open ticket already exists for this issue"

If NO → continue

Validation 2: Vehicle Status

Is vehicle Decommissioned or Archived? If YES → blocked. If NO → proceed.

System creates ticket, assigns Custodian, sends notification

Edge Case: Duplicate Ticket Attempt

Paul creates Ticket 1 for Firetruck FT-02: "Engine oil change". Later, he tries to create Ticket 2 with identical description. System blocks with: "Open ticket for this vehicle with matching title already exists on 2026-07-20."

Edge Case: Ticket on Decommissioned Vehicle

Vehicle TV-99 is marked Decommissioned. Paul tries to create a ticket. System blocks: "Cannot create ticket on decommissioned vehicle." Vehicle must be restored (status changed back to active) before new tickets can be created.

5. Custodian Inspects Main Issue

After ticket is created, the assigned Custodian physically inspects the vehicle to verify the Main Issue is real. Submits verdict: No Issues, Needs Maintenance, or Not Fit for Service. This is Tier-1 verification.

SCENARIO: CUSTODIAN VERIFIES ISSUE - NEEDS MAINTENANCE

Sarah (Custodian) Custodian receives ticket TK-2026-1847 for brake inspection. Goes to garage, inspects Ambulance AMB-101 brakes: feels brake pedal (soft), visually checks brake pads (worn nearly to backing), checks brake fluid level (low by 1 inch). Confirms: this is a real problem needing maintenance.

Sarah (Custodian) Custodian in ticket form, selects verdict = "Needs Maintenance". Checks attestation: "I have personally inspected this vehicle". Creates sub-issue: "Replace brake pads - wear/replacement". Submits inspection.

System updates ticket status: Open → Active (now in active repair phase). Creates sub-issue (status=Open). Notifies Admin that inspection is complete.

Paul (Admin) Admin sees ticket now Active. Reviews Sarah's inspection findings and sub-issue. Decides to assign James (Mechanic) to repair the brake pads.

SCENARIO: CUSTODIAN FINDS NO ISSUES (FALSE ALARM)

Lisa (Custodian) Custodian receives ticket for "Engine overheating" on Firetruck FT-02. Inspects: temperature gauge reads normal (180°F typical), no coolant leaks, radiator hoses intact, thermostat opens/closes normally. Everything appears normal — this is likely a false alarm or sensor glitch.

Lisa (Custodian) Custodian submits verdict = "No Issues". Attestation: "I have personally inspected this vehicle. Temperature and cooling system appear normal."

No sub-issues created.

System if no sub-issues created, ticket jumps directly to Admin confirmation phase (no repair needed).

Paul (Admin) Admin reviews Lisa's inspection. Agrees — no work needed. Approves ticket, marks it Closed. Issue Report marked Resolved.

EXPECTED OUTCOMES

- ✓ Ticket status: Open → Active (if Needs Maintenance) or skips to Closed (if No Issues)
- ✓ Sub-issues created only if Needs Maintenance verdict
- ✓ Custodian attestation required ("I have inspected")
- ✓ Vehicle history updated with inspection record
- ✓ Admin notified of inspection results
- ✓ If No Issues: ticket closes without repair
- ✓ If Needs Maintenance: ticket enters repair phase

DATA FLOW & DECISIONS

Custodian opens assigned ticket, inspects vehicle physically, submits verdict + attestation

Decision: Inspection Verdict

Verdict = "No Issues" → no sub-issues, ticket jumps to Admin confirmation, then closes

> Verdict = "Needs Maintenance" → create sub-issues, ticket status = Active, proceed to mechanic assignment

> Verdict = "Not Fit for Service" → create sub-issues, vehicle status auto-changes to Out of Service

Edge Case: Inspection Without Attestation
Sarah clicks submit but forgets to check the attestation box. System blocks submission with error: "Attestation required. Please confirm you have personally inspected this vehicle."

Edge Case: Multiple Sub-Issues From Single Inspection
During brake inspection, Sarah finds: (1) worn brake pads, (2) low brake fluid, (3) soft brake pedal. She creates three separate sub-issues. Ticket now has 3 tasks, each will be assigned separately.

6. Assign Mechanic to Sub-Issue

After Custodian verifies the Main Issue, Admin assigns a Mechanic (Maintenance Personnel) to each sub-issue. This creates a work order. Mechanic receives notification and can begin work.

SCENARIO: ADMIN ASSIGNS MECHANIC

Paul (Admin) Admin ticket TK-2026-1847 now has sub-issue: "Replace brake pads". Paul opens sub-issue details, clicks "Assign Mechanic", searches for available mechanics. Selects James (Maintenance Personnel). Notes: "Standard replacement per manual. Use OEM pads."

System updates sub-issue: assigned_mechanic=James, status=Open (waiting for mechanic to accept/start). Creates work order. Sends notification to James: "You've been assigned sub-issue #1 of Ticket TK-2026-1847: Replace brake pads (AMB-101)".

James (Maintenance Personnel) Mechanic receives notification. Opens sub-issue, reviews details. Notes parts needed (OEM brake pads, part #BP-2426), estimated time (2 hours). Marks sub-issue "Under Repair" to indicate work has begun.

System updates sub-issue status: Open → Under Repair. Vehicle status may auto-change to Under Maintenance if not already.

EXPECTED OUTCOMES

- ✓ Sub-issue assigned to specific Mechanic
- ✓ Work order created with description and notes
- ✓ Mechanic notified of assignment
- ✓ Sub-issue ready for mechanic to begin work
- ✓ Vehicle status may auto-update to Under Maintenance if necessary

✓ Assignment tracked in audit history

DATA FLOW & VALIDATION

Admin opens Sub-Issue, clicks "Assign Mechanic", selects from list of available Maintenance Personnel

Admin optionally adds notes (specifications, part numbers, warnings)

Validation

Is Mechanic already assigned to open sub-issues? (optional soft limit, not enforced by system)

System creates work order, notifies Mechanic, updates history

Edge Case: No Mechanics Available
All mechanics are assigned to other tickets. Admin can still assign a mechanic (system does not enforce max workload limits), but the sub-issue may remain Open longer until mechanic has capacity.

Edge Case: Multiple Sub-Issues, Different Mechanics
Ticket has 3 sub-issues: (1) Replace brake pads (assigned James), (2) Check fuel system (assigned David), (3) Inspect lights (assigned James again). Each mechanic gets their assignments separately.

7. Mechanic Logs Repair Work

Mechanic performs the actual repair, logs details: parts used, hours spent, date completed, cost. Marks sub-issue "For Inspection" when ready for Custodian verification.

SCENARIO: MECHANIC REPLACES BRAKE PADS

James (Maintenance Personnel) Mechanic assigned sub-issue #1: Replace brake pads (AMB-101). Orders OEM brake pads (part #BP-2426, cost $85). Pads arrive next morning. James removes wheels, inspects brake system, removes old pads, installs new pads, bleeds air from lines, test-drives to verify firm brake response.

James (Maintenance Personnel) Mechanic logs repair work in sub-issue form: date_completed=2026-07-21, hours_spent=2.0, parts_used=["OEM Brake Pads #BP-2426"], parts_cost=$85, labor_cost=$0 (internal labor), total_cost=$85, notes="Replaced front and rear brake pads per OEM spec. Bled brake lines. Tested responsiveness — firm and proportional. Ready for verification."

System saves work log, updates sub-issue status: Under Repair → For Inspection. Sends notification to assigned Custodian (Sarah): "Sub-issue #1 of Ticket TK-2026-1847 is ready for your inspection."

EXPECTED OUTCOMES

- ✓ Work log recorded with detailed repair information
- ✓ Parts used and costs documented
- ✓ Sub-issue status: Under Repair → For Inspection
- ✓ Custodian notified that repair is ready for verification
- ✓ Work history preserved for audit and future reference

DATA FLOW

Mechanic opens assigned sub-issue, performs repair, logs work details (parts, cost, hours, notes)

System saves log, updates sub-issue status to For Inspection

System notifies assigned Custodian: sub-issue ready for verification

Edge Case: Mechanic Discovers Additional Issues
While replacing brake pads, James notices the master cylinder is also leaking. He completes the pads replacement, notes the leak in his log, and suggests creating a new sub-issue for master cylinder repair.

8. Custodian Verifies Repair

After mechanic completes work, Custodian test-drives/inspects to confirm repair is effective. Submits verdict: Approved or Rejected. This is quality gate before final Admin confirmation.

SCENARIO: VERIFICATION APPROVED

Sarah (Custodian) Custodian receives notification sub-issue #1 ready for verification. Test-drives AMB-101, applies brakes at 10mph, 25mph, 50mph. All responses firm and proportional. Feels confident repair is effective. Submits: verdict=Approved, attestation="I have personally tested this vehicle", comments="Brake response is firm and consistent. Repair meets specifications."

System updates sub-issue status: For Inspection → For Confirmation. Notifies Admin (Paul) that sub-issue passed inspection and is ready for final confirmation.

SCENARIO: VERIFICATION REJECTED - REWORK NEEDED

Lisa (Custodian) Custodian assigned to verify transmission rebuild on Firetruck FT-02. Test-drives, shifts through all gears. Finds shift timing is sluggish (0.8 second delay between gears). This exceeds OEM spec (should be <0.3 second). Submits: verdict=Rejected, reason="Shift timing out of spec. Requires recalibration", comments="Shifts are functional but timing is delayed. Does not meet OEM specification."

System updates sub-issue status: For Inspection → Under Repair (returns to mechanic). Notifies mechanic (David) that repair was rejected and requires rework. Includes Custodian's reason: "Shift timing out of spec."

David (Maintenance Personnel) Mechanic receives rejection notification. Reviews reason. Recalibrates transmission shift timing per OEM spec. Test-drives internally to verify shift timing is now <0.3 second. Marks sub-issue For Inspection again.

Lisa (Custodian) Custodian receives notification for re-verification. Test-drives again. Shift timing is now <0.2 second. Approves. Sub-issue moves to For Confirmation.

EXPECTED OUTCOMES

- ✓ Custodian test-drives/inspects repaired vehicle
- ✓ Submits verdict: Approved or Rejected
- ✓ If Approved: sub-issue status For Inspection → For Confirmation (awaiting Admin)
- ✓ If Rejected: sub-issue status For Inspection → Under Repair (mechanic reworks)
- ✓ Rejection reason captured for mechanic to address
- ✓ Attestation required

DATA FLOW & DECISIONS

Custodian receives notification sub-issue ready for verification, test-drives/inspects, submits verdict + attestation

Decision: Verification Verdict

Verdict = Approved → status For Inspection → For Confirmation

Verdict = Rejected → status For Inspection → Under Repair, include rejection reason

9. Admin Confirms Repair

Final approval step (Tier-2 Confirmation). Admin reviews Custodian's verification and mechanic's work. Marks sub-issue Done or Rejects (returns to Under Repair).

SCENARIO: ADMIN CONFIRMS - REPAIR APPROVED

Paul (Admin) Admin sees sub-issue #1 awaiting final confirmation. Reviews: (1) James's work log (parts, hours, cost), (2) Sarah's verification notes ("Brake response firm"), (3) parts receipt. All validates. Approves: verdict=Confirmed_Done.

System updates sub-issue status: For Confirmation → Done. If all sub-issues are Done, ticket status: Active → Closed. Creates Maintenance Record (immutable): ticket_id=TK-2026-1847, vehicle=AMB-101, work="Replaced brake pads", mechanic=James, custodian=Sarah, cost=$85, date_completed=2026-07-21.

System updates vehicle status: Under Maintenance → Available. Marks Issue Report Resolved. Sends notifications to all three users: Sarah, James, Paul.

EXPECTED OUTCOMES

- ✓ Admin reviews work and verification
- ✓ Sub-issue status: For Confirmation → Done (or Reopen to Under Repair)
- ✓ If all sub-issues Done: Ticket closes, Maintenance Record created
- ✓ Vehicle status updated back to Available
- ✓ Issue Report marked Resolved
- ✓ All parties notified

10. Defer a Sub-Issue

Instead of completing a sub-issue, mechanic or admin marks it Deferred (work postponed). System auto-creates Issue Report breadcrumb so deferred work is not forgotten.

SCENARIO: PARTS BACKORDERED - DEFER WORK

James (Maintenance Personnel) Mechanic assigned to replace transmission on Vehicle TV-08. Discovers transmission part #TX-5500 is backordered until 2026-09-15 (8 weeks away). Vehicle is urgently needed for mission next week. James marks sub-issue status: Deferred, reason="Transmission part #TX-5500 backordered. ETA 2026-09-15".

System updates sub-issue status: Under Repair → Deferred. Auto-creates Issue Report: title="Vehicle TV-08 pending transmission replacement", description="Transmission part #TX-5500 backordered. ETA 2026-09-15. Work to resume upon part arrival", status=Requires Follow-up.

Paul (Admin) Admin reviews the deferred work. Decides to close the ticket as Decision Close (work incomplete, intentionally deferred). Vehicle can be dispatched with understanding that it will return for transmission replacement.

System closes ticket status: Active → Closed (Decision Close result). Creates Maintenance Record with note: "Work deferred - part backordered". Issue Report remains open in Admin's queue as reminder.

2026-09-15 (Future) Part arrives. Paul reviews the Issue Report, creates new Ticket #2 to complete the transmission replacement. James performs the work.

EXPECTED OUTCOMES

✓ Sub-issue status: Under Repair → Deferred

- ✓ Auto-created Issue Report with deferred work description
- ✓ Ticket can be closed as Decision Close with incomplete work
- ✓ Deferred work tracked and visible to Admin
- ✓ Future follow-up can be initiated when conditions permit
- ✓ No silent loss of work — breadcrumb ensures follow-up

11. Close Ticket (Complete/Done)

All sub-issues are Done. Admin approves final closure. Ticket moves to Closed status (result=Done). Maintenance Record created. Vehicle released. Issue Report resolved.

SCENARIO: ALL SUB-ISSUES COMPLETE - TICKET CLOSES

Paul (Admin) Admin ticket TK-2026-1847 now has all sub-issues marked Done. Paul reviews overall ticket completion: all work performed, all verifications passed, all costs logged. Approves final closure: verdict=Closed_Done.

System ticket status: Active → Closed (result=Done). Creates Maintenance Record (immutable). Updates vehicle status: Under Maintenance → Available. Marks Issue Report Resolved. Adds entries to vehicle history and ticket audit log.

Dashboard vehicle AMB-101 now shows: status=Available, condition=Good, readiness=Ready (if recent check exists). Vehicle can be dispatched immediately.

EXPECTED OUTCOMES

- ✓ Ticket status: Active → Closed (Done)
- ✓ All sub-issues status: Done
- ✓ Maintenance Record created and locked (immutable)
- ✓ Vehicle status: Under Maintenance → Available
- ✓ Issue Report status: Pending → Resolved
- ✓ All notifications sent

12. Close Ticket (Decision Close)

Work is incomplete, but ticket is intentionally closed anyway (e.g., parts backordered, vehicle decommissioned, work deferred). Auto-creates Issue Report breadcrumb for future follow-up.

SCENARIO: ENGINE OVERHAUL INCOMPLETE - DEFER AND CLOSE

Paul (Admin) Admin ticket TK-2026-1850: Engine overhaul (Vehicle TV-12). Mechanic James has completed 40% of work. Overhaul was originally estimated 3 weeks; it's been 4 weeks and will take another 3-4 weeks. Vehicle is urgently needed. Paul decides to pause the overhaul, dispatch the vehicle, and resume when it returns. Marks sub-issue: Deferred, then closes ticket: result=Decision_Close, reason="Engine overhaul paused — vehicle needed for urgent mission. Work to resume upon return."

System marks sub-issue Deferred. Auto-creates Issue Report: "Vehicle TV-12 engine overhaul on hold — 60% incomplete. ETA for completion TBD". Closes ticket status: Active → Closed (Decision Close). Vehicle status: Under Maintenance → Available (despite incomplete work).

Paul (Admin) Admin vehicle TV-12 is now available for dispatch. Dashboard shows: Status=Available, Condition=Needs Maintenance (from earlier check), Issue Report=Open (pending engine overhaul). Acknowledges risk, but operational need overrides.

EXPECTED OUTCOMES

- ✓ Ticket status: Active → Closed (Decision Close)
- ✓ Sub-issues marked Deferred (not Done)
- ✓ Auto-created Issue Report for deferred work
- ✓ Vehicle status: Under Maintenance → Available (despite incomplete work)

- ✓ Maintenance Record created with note about Decision Close
- ✓ Deferred work tracked for future follow-up

13. Reassign Mechanic Mid-Repair

If assigned mechanic becomes unavailable (sick, transferred, etc.), Admin reassigns ticket/sub-issue to another mechanic. Work history preserved; no loss of prior progress.

SCENARIO: MECHANIC GETS SICK - REASSIGNMENT

James (Maintenance Personnel) Mechanic assigned to sub-issue: transmission rebuild (Firetruck FT-02). Work is 60% complete (Under Repair status). James falls ill, will be out for 2 weeks.

Paul (Admin) Admin notified James is unavailable. Opens sub-issue, reassigns to David (another mechanic). Notes: "James was 60% complete on transmission rebuild. Refer to his work log for progress notes."

System updates sub-issue: assigned_mechanic = James → David. Sends notification to David: "You've been assigned sub-issue #1 (in progress) of Ticket #XYZ: Transmission Rebuild (FT-02). Mechanic James was 60% complete. See work log for details."

David (Maintenance Personnel) Mechanic reviews sub-issue, reads James's work log (what's been done, what remains), reviews any notes or warnings. Continues work from 60% complete to finish. Completes rebuild, logs his own work additions, marks For Inspection.

System work history shows both James and David worked on this sub-issue. Audit trail shows: James worked Day 1-10, David worked Day 11-15. Timeline preserved.

EXPECTED OUTCOMES

- ✓ Sub-issue reassigned to new mechanic
- ✓ Work history and progress notes preserved

- ✓ New mechanic can see prior work log and continue
- ✓ Audit trail shows both mechanics involved
- ✓ No loss of progress or prior work

14. Create Preventive Maintenance Schedule

Admin or Maintenance Personnel schedules recurring maintenance ahead of time (e.g. "Oil change every 6 months") instead of relying on someone to remember. A schedule with a recurrence set automatically regenerates its next occurrence once completed (see Workflow 15).

SCENARIO: RECURRING OIL CHANGE SCHEDULE

Michael (Admin) Admin opens Maintenance Schedule, clicks "Add". Fills: vehicle=Ambulance AMB-101, maintenance_type="Oil Change", scheduled_date=2026-08-15, scheduled_time=09:00, service_location="Barangay Motor Pool", assigned_to=James (Maintenance Personnel), recurrence_months=6 ("Every 6 mo").

System checks: does AMB-101 already have a Scheduled entry on 2026-08-15? No → validation passes. Creates schedule with status=Scheduled.

James (Maintenance Personnel) Mechanic sees the new entry in Maintenance Schedule, assigned to him, due 2026-08-15.

EXPECTED OUTCOMES

- ✓ Schedule created with status = Scheduled
- ✓ Recurrence is stored as a raw month count (1–60), not limited to the four labeled presets
- ✓ Schedule appears in the Maintenance Schedule list, assignable to a specific mechanic
- ✓ Duplicate same-day schedules on the same vehicle are blocked

DATA FLOW & VALIDATION

Admin / Mechanic Maintenance Schedule → Add: vehicle, maintenance_type, scheduled_date, scheduled_time, service_location, assigned_to, recurrence_months

Validation: Duplicate Same-Day Schedule

Does this vehicle already have a Scheduled entry on the exact same date? If YES → blocked. If NO → proceed.

System creates schedule, status = Scheduled

Edge Case: Duplicate-Date Schedule
Michael tries to add a second schedule for AMB-101 also dated 2026-08-15 (e.g. accidentally submitted twice). System blocks with: "This vehicle already has a maintenance schedule on that date."

Edge Case: Non-Standard Recurrence
Michael sets recurrence_months=4, which isn't one of the four labeled presets (Monthly/Quarterly/6 mo/Yearly). The backend accepts any value 1–60; the UI simply displays it as "Every 4 months" instead of a named label.

15. Complete Scheduled Maintenance

When the scheduled date arrives, the assigned mechanic (or Admin) marks it done. Unlike a ticket-based repair, this reaches "Completed" immediately — no separate Custodian verify / Admin confirm cycle — and if the schedule carries a recurrence, the next occurrence is auto-created on the spot.

SCENARIO: MARK DONE, RECURRENCE AUTO-REGENERATES

James (Maintenance Personnel) Mechanic performs the scheduled oil change on AMB-101. Opens Maintenance Schedule, clicks "Mark Done" on today's entry. Fills: date_completed=today, maintenance_cost=850, notes="Standard oil + filter change, no issues found."

System checks schedule.status === Scheduled → passes. Creates a VehicleMaintenanceRecord (maintenance_type="Oil Change", problem_reason="Scheduled preventive maintenance", progress_status=Completed immediately, remarks="Completed from schedule #1204."). Updates the schedule itself to status=Completed.

System since recurrence_months=6 was set on this schedule, auto-creates a brand-new schedule: scheduled_date = today + 6 months, same maintenance_type/time/location/assigned_to/recurrence_months, status=Scheduled.

Michael (Admin) Admin sees a completed Maintenance Record for the oil change (no further verification needed) and a fresh Scheduled entry already queued 6 months out.

EXPECTED OUTCOMES

✓ A VehicleMaintenanceRecord is created with progress_status = Completed immediately

- ✓ The schedule itself moves to status = Completed
- ✓ If recurrence_months is set, the next schedule is created automatically in the same transaction
- ✓ If no recurrence is set, nothing further is auto-created

DATA FLOW & DECISIONS

Mechanic / Admin Maintenance Schedule → Mark Done: date_completed, maintenance_personnel_id, maintenance_cost, notes

Precondition: Schedule Status

schedule.status must be Scheduled. Any other status → blocked with the current status shown in the error.

Decision: Recurrence Set?

recurrence_months present → auto-create next schedule (scheduled_date + N months). recurrence_months null → one-time schedule, nothing further created.

Edge Case: Marking an Already-Completed Schedule Done
James accidentally clicks "Mark Done" a second time on the same schedule (already Completed). System blocks with: "Only a scheduled maintenance can be marked done. Current: Completed."

Edge Case: One-Time Schedule, No Recurrence
Lisa completes a one-off "Windshield replacement" schedule with recurrence_months

left blank. The response's `next` field is null — no follow-up schedule is generated, since this was never meant to repeat.

16. Add Maintenance Record (Standalone)

For work that never goes through the ticket pipeline — an external vendor repair, or field work discovered outside the normal issue-report flow — Admin or Maintenance Personnel can log a Maintenance Record directly. Reaching "Completed" on it, however, still requires the same Custodian-verify → Admin-confirm gate used elsewhere; a standalone record cannot be marked Completed by direct edit.

SCENARIO: EXTERNAL VENDOR REPAIR, LOGGED AFTER THE FACT

Paul (Admin) Admin Firetruck FT-02 was sent to an external vendor (Naga Motorworks) for a specialized pump-seal repair — no VMS ticket was ever opened for it. Paul opens Maintenance Records, clicks "Add". Fills: vehicle=FT-02, maintenance_type="Engine Repair", problem_reason="Pump seal failure - repaired by external vendor Naga Motorworks", action_taken="Vendor replaced pump seal and pressure tested", parts_used="Pump seal kit", maintenance_cost=4500. progress_status left blank.

System validates required fields, defaults progress_status to "Assigned" since none was supplied. As a side effect of any new maintenance record, FT-02's vehicle status is forced to "Under Maintenance" and condition to "Needs Repair" — even though the vendor work is already physically done.

Lisa (Custodian) Custodian sees the record in her Maintenance Status queue, inspects the pump herself, confirms it holds pressure with no leak. Submits verification_result = Passed.

Paul (Admin) Admin now that verification has Passed, updates progress_status to "Completed". The record is finalized in the vehicle's maintenance history.

EXPECTED OUTCOMES

- ✓ Maintenance Record created directly, with no ticket involved
- ✓ Vehicle status/condition forced to Under Maintenance / Needs Repair the moment the record is created
- ✓ If the creator is Maintenance Personnel (not Admin), maintenance_personnel_id is forced to their own id regardless of what was submitted
- ✓ progress_status cannot reach Completed except through Custodian-verify → Admin-confirm

DATA FLOW & VALIDATION

Admin / Mechanic Maintenance Records → Add: vehicle_id, maintenance_type, problem_reason (required); action_taken, parts_used, maintenance_cost, progress_status (optional)

Rule: Personnel Assignment

Caller is Maintenance Personnel only (not also Admin) → maintenance_personnel_id is overwritten to their own id. If somehow still empty → blocked: "Please select maintenance personnel."

Rule: Reaching Completed

Direct edit to progress_status=Completed without a Passed verification_result → blocked. Only allowed once Admin confirms after a Custodian's Passed verification.

Edge Case: Skipping Straight to Completed

Paul tries to set progress_status="Completed" directly when creating the record (no verification has happened yet). System blocks with: "A maintenance record can only be completed by an Admin after Custodian verification has passed. Use the verify/confirm workflow instead."

Edge Case: No Restriction Against an Active Ticket
Unlike Decommission (Workflow 17), creating a standalone Maintenance Record has no check against existing open tickets on the same vehicle — a vehicle can have both an open ticket AND a standalone record in flight at once.

17. Decommission a Vehicle

When a vehicle reaches end of life (too old, uneconomical to repair), Admin permanently retires it. Its full history is preserved, but it's excluded from every readiness/coverage/fragility calculation from that point on. The action is reversible — Admin can recommission it later if it turns out to be a mistake, or funding appears for a full refurbishment.

SCENARIO: RETIRING AN AGING FIRE TRUCK

Paul (Admin) Admin Firetruck FT-01 is 22 years old; its chassis has rusted beyond safe repair. Paul opens its vehicle profile and clicks "Decommission".

System checks for open tickets on FT-01 (status not Closed/Cancelled). Finds an Active ticket, "Pump pressure loss", still in progress. Blocks with: "Close or cancel this vehicle's open ticket(s) before decommissioning it."

Paul (Admin) Admin opens that ticket and closes it as a Decision Close (Workflow 12) — no point completing the repair on a vehicle about to be retired. Returns to the vehicle profile and retries Decommission.

Paul (Admin) Admin no open tickets remain now → proceeds. Enters decommission_reason: "22 years old, chassis rust exceeds safe repair threshold. Approved for retirement by barangay council resolution #2026-014." Submits.

System sets status=Decommissioned, condition=Damaged, estimated_return_date=null, plus decommission_reason, decommissioned_by=Paul, decommissioned_at=now. Logs a "Vehicle Decommissioned" entry to Vehicle History. FT-01 immediately drops out of the Fire Truck category's readiness/coverage/fragility numbers on the Dashboard.

Paul (Admin) Admin a month later, the barangay unexpectedly secures funding to fully refurbish FT-01. Paul reopens its profile and clicks "Recommission". System resets status=Available, condition=Good, clears the decommission fields, and logs it as "recommissioned" in Vehicle History.

EXPECTED OUTCOMES

- ✓ Vehicle excluded from every readiness/coverage/fragility calculation while Decommissioned
- ✓ Full history preserved; decommission_reason/by/at retained until recommissioned
- ✓ Any open ticket on the vehicle blocks decommissioning until closed or cancelled
- ✓ Fully reversible — Recommission restores Available/Good and clears the decommission fields

DATA FLOW & VALIDATION

Admin Vehicle Profile → Decommission: decommission_reason (required)

Validation 1: Already Decommissioned?

vehicle.status already Decommissioned → blocked. Else → continue.

Validation 2: Open Tickets

Any MaintenanceTicket on this vehicle not Closed/Cancelled → blocked. Else → proceed.

System sets status/condition, stamps decommission fields, writes history

Edge Case: Decommissioning an Already-Decommissioned Vehicle

Paul mistakenly clicks Decommission again on FT-01 (already Decommissioned). System blocks with: "This vehicle is already decommissioned."

Edge Case: Open Ticket Blocks Decommission

Michael tries to decommission Ambulance AMB-101 while it still has an Active ticket. System blocks with: "Close or cancel this vehicle's open ticket(s) before decommissioning it." — decommissioning is a dead end for a vehicle mid-repair unless that work is explicitly closed or cancelled first.

18. Add Sub-Issue Mid-Ticket

While repairing one problem, a mechanic or custodian sometimes discovers another, unrelated root cause on the same vehicle. Rather than opening a whole new ticket, they can add a new sub-issue directly onto the current Active ticket — but only if they are the actual assigned Custodian or one of the assigned Mechanics on it. Admin is explicitly excluded from this action, even on an Admin+Custodian combined-role account — declaring a new root cause requires firsthand contact with the vehicle, the same principle behind the Custodian-only initial inspection (Workflow 5).

SCENARIO: SECOND PROBLEM FOUND MID-REPAIR

James (Maintenance Personnel) Mechanic is the assigned mechanic on sub-issue "Brake pads" of Ticket #1847 (Active). While under the ambulance, he notices the rear differential is also leaking oil — an unrelated problem no one had reported.

James (Maintenance Personnel) Mechanic cannot open a brand-new ticket himself (ticket creation is Admin-only, Workflow 4). Instead opens Ticket #1847, clicks "Add Sub-Issue". Title: "Rear differential oil leak - discovered during brake repair". Leaves maintenance_type blank for Admin to assign.

System checks ticket.status === Active → passes. Checks James is an assigned mechanic on an existing sub-issue of this ticket → ownership check passes. Creates the new sub-issue with status=Open. Ticket progress recalculates from 0/1 done to 0/2 — nothing is reset, the total simply grew.

Paul (Admin) Admin sees Ticket #1847 now carries 2 sub-issues, one still Open (the differential leak). Assigns it a maintenance_type and a mechanic through the normal Assign Mechanic step (Workflow 6).

EXPECTED OUTCOMES

- ✓ New sub-issue runs its own repair → verify → confirm pipeline, independent of the original one
- ✓ Ticket stays Active until every sub-issue (old and new) reaches Done or Deferred
- ✓ Ownership is checked by literal user id, not just role — a Mechanic on a different ticket cannot add here even sharing the same role
- ✓ Admin can never add a sub-issue directly, regardless of any other role they hold

DATA FLOW & VALIDATION

Custodian / Mechanic Ticket Detail → Add Sub-Issue: title (required); maintenance_type, issue_report_id (optional)

Validation 1: Ticket Status

ticket.status must be Active. Any other status → blocked, current status shown in the error.

Validation 2: Ownership

Caller must be the ticket's assigned_custodian_id, OR have an existing sub-issue on this ticket assigned to them as mechanic. Neither → blocked 403.

Edge Case: Uninvolved Custodian Tries to Add a Sub-Issue
Precious (Custodian), who is NOT the assigned custodian on Ticket #1847 (that's Sarah), opens the ticket and tries to add a sub-issue anyway. System blocks with: "You are not currently assigned to this ticket."

Edge Case: Ticket No Longer Active
David tries to add a sub-issue to a ticket that has already reached For Confirmation.

System blocks with: "Sub-issues can only be added while the ticket is Active. Current status: For Confirmation."

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All 18 workflows documented with detailed scenarios, decision points, and edge cases. Professional staff names used throughout (Paul, Sarah, James, Lisa, Michael, David, etc.).